Multimodal Fashion & Context Retrieval

Compositional text-to-image retrieval over a fashion corpus

1. The problem

CLIP is a strong zero-shot retrieval baseline and a weak compositional one. A dual encoder pools an image into a single vector, so **red shirt + blue pants** and **blue shirt + red pants** land in nearly the same place: the model registers that {red, blue, shirt, pants} are present but loses **which colour attaches to which garment**. This bag-of-words behaviour is well documented in contrastive vision-language models, and it is the failure the brief's hint points at.

Every design decision below exists to fix attribute binding. The system is measured on whether it can tell a query from its colour-swapped twin — a property vanilla CLIP largely does not have.

2. Approaches considered

Five viable architectures, in rough order of cost:

Approach	How it works	Good when	Weakness
A. Vanilla CLIP, global vector	One embedding per image; cos[illegible]	Fast to build, genuinely zero-shot, scales trivially. [illegible]	No attribute binding. [illegible]
B. Domain-tuned backbone	Swap in a fashion-fine-tuned encoder (FashionCLIP, FashionSigLIP) [illegible]	[illegible]	Still a dual encoder [illegible]
C. Region multi-vector (chosen)	Embed each garment crop sep[illegible]	Directly attacks binding [illegible]	[illegible]
D. Cross-encoder rerank	Joint image-text encoder (BLIP-ITM) [illegible]	Cross-attention resolves binding [illegible]	[illegible] — cannot touch the [illegible]
E. Fine-tune with hard negatives	Train on colour-swapped nega[illegible]	Best ceiling; pushes binding into the embedding [illegible]	Needs labelled attribute data (absent here), GPU budge[illegible]

Chosen: B + C, with D scoped as the next increment. B is free. C is where the brief's stated failure mode actually lives. D is deliberately deferred rather than dropped — it is the strongest precision lever left, and the shortlist architecture is already built to accept it. E is out of reach: the dataset ships no attribute labels to train on (see §4).

3. Chosen architecture

Modelled on **LookSync**, Glance's production visual product search system (arXiv:2511.00072): decompose the look into layer-wise garment descriptions → vectorise → vector-DB retrieval → rerank. The same four stages, adapted from *product retrieval given an AI-generated look* to *scene retrieval given natural language*.

Notably, LookSync's own bake-off put vanilla CLIP ViT-H/14 *above* FashionCLIP and Fashion-SigLIP, by 3–7% MOS — on catalogue product retrieval. That result did not transfer here, which is why the backbone was chosen by measurement rather than assumption (§5).

Part A — Indexer

Two vector kinds per image:

Vector	Source	Answers
global	whole frame	scene ('inside a modern office'), vibe ('casual weekend')
region	one per garment box	which colour belongs to which garment

Region boxes are filtered to the 27 real garment/accessory categories. **46% of Fashionpedia's boxes are parts** — sleeve, neckline, zipper, rivet — and a crop of a zipper is index noise. There is deliberately no third 'scene vector': with a single backbone that is just the global vector under another name. Storage is ChromaDB, per the brief's instruction to pick a convenient store rather than build one; the ML logic is in *what* is embedded, not the store.

Part B — Retriever

Decomposition is zero-shot by construction. The brief grades handling of 'descriptions it hasn't seen explicitly in a training label', which rules out the obvious implementation. A parser with a hardcoded colour list and garment taxonomy passes all five graded queries and silently drops any term outside its vocabulary — closed-vocabulary retrieval wearing a zero-shot costume. So no colours or garments are ever enumerated:

- **Split** on conjunctions and commas — purely syntactic, no word list.
- **Route** each clause by comparing it against *anchor prompts* describing what a garment is versus what a place is, using the text encoder itself. The encoder knows 'windbreaker' even though we never wrote it down. Unsure clauses fall back to the global vector, so the system degrades to CLIP rather than to nothing.

Binding via one-to-one assignment. Region scoring alone is *not* enough, and this is the subtle part. If each clause independently takes its best-matching region, then for an image with a **red shirt** and a **white tie**, the clause 'a red tie' happily matches the red *shirt* crop — colour similarity dominates garment identity, the swapped image still scores well, and nothing has been fixed.

Instead garment clauses are assigned to regions **one-to-one**, greedily by descending similarity. Clauses compete: once 'a red tie' claims the red crop, 'a white shirt' cannot also have it. An image genuinely containing a red tie and a white shirt satisfies both clauses with two different crops; the colour-swapped image satisfies neither well. This needs no category lookup, so it remains open-vocabulary.

Fusion z-normalises per clause. Raw scores are not comparable across clauses or backbones — SigLIP's pairwise sigmoid loss puts its cosines on a completely different scale from CLIP's InfoNCE (a random image scores +0.18 under CLIP, –0.05 under SigLIP) — so absolute thresholds would be meaningless.

4. Dataset: built, not sampled

1,788 images = 1,200 Fashionpedia + 588 COCO. Two audits drove this, and both changed the design.

Audit 1 — Fashionpedia's distribution starves the graded queries

tie appears in 3 of 1,158 val images. The compositional query — the one the entire architecture exists to win — would have had essentially no ground truth. Train is 39× larger and holds 1,455 tie images and 1,402 tie+shirt images, so the corpus is **stratified from train with a quota per graded query** (200 tie+shirt, 200 formal, 180 outerwear, 160 shirt, 200 casual, 260 random distractors), filled rarest-first so a tie+shirt image is never consumed by the distractor bucket.

Audit 2 — Fashionpedia cannot serve the environment axis

Zero-shot scene tagging over the sample found it is **85% red carpet, runway and studio**. The four environments the brief requires total **14.2%** — 19 office images and 4 home images out of 1,200 — while three of the five graded queries are environment-driven.

Scene	Share
red carpet	36.2%
runway / catwalk	29.8%
studio backdrop	19.1%
urban street / park / office / home	14.2% ← the required axis

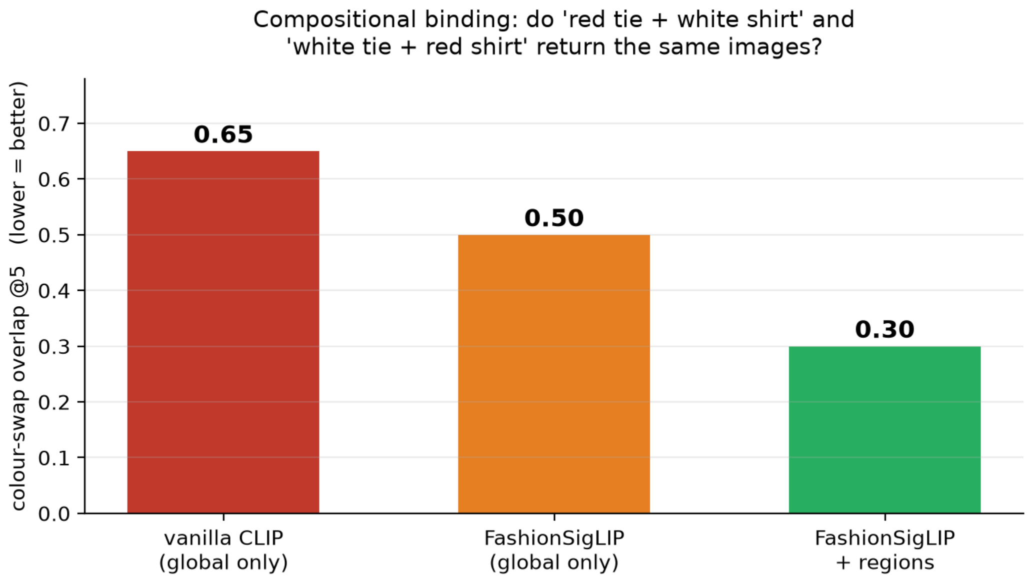
COCO fixes that, and does one thing a generic scene dataset could not: **its object labels give scene ground truth that is not derived from CLIP**. A 'bench' box is a human annotation, so grading a CLIP retriever against it is not circular — which CLIP-generated scene tags could never be. Scene is inferred from co-present objects with person always required: office ← laptop/keyboard/mouse (89), home ← couch/bed/tv (175), park ← bench (149), street ← car/traffic light/bus (175).

Note the honest limit: COCO rows carry no garment annotations, so they contribute a global vector only. They are scene evidence and distractors, not garment evidence.

5. Results

The metric. Fashionpedia's HF release ships no colour or scene attributes, so there are no labels to compute Precision@5 against. LookSync hit the same wall and fell back to human Mean Opinion Score. So the headline metric needs no labels:

Retrieve top-5 for *'a red tie and a white shirt'*, then for *'a white tie and a red shirt'*. Both queries contain an **identical bag of words**. A system that ignores binding returns nearly the same images for both (overlap → 1.0). A system that binds returns different ones (overlap → 0). **Lower is better.** The metric is rank-based (no score comparability needed) and self-supervised (no ground truth needed): it measures *sensitivity to binding*, exactly the property the brief's hint calls out. Averaged over 4 swap pairs, k=5.



Configuration	Swap overlap ↓	Reading
vanilla CLIP (global only)	0.65	Returns 65% of the same images either way — it cannot tell the queries apart.
FashionSigLIP (global only)	0.50	Domain vocabulary helps, but pooling still destroys binding.
FashionSigLIP + regions	0.30	Region scoring + assignment does the heavy lifting: 0.50 → 0.30.

The backbone swap buys 0.65 → 0.50; **the mechanism buys 0.50 → 0.30**. That split matters: the gain is attributable to the architecture, not to having picked a fancier model. Zero-shot decomposition routes correctly with no vocabulary — 'A red tie' → garment, 'a white shirt' → garment, 'in a formal setting' → scene.

Known limitations

- **Query 4 fails to decompose.** 'Casual weekend outfit for a city walk' routes as a single garment clause — the syntactic splitter has no rule for 'for a…', so 'city walk' never reaches the scene vector and results skew editorial rather than street. An LLM decomposer (LookSync's actual choice) would handle this; the syntactic splitter is the deliberate trade for zero API dependency.
- **4 swap pairs is a small sample.** The ordering is consistent and the gap is wide, but the absolute values carry meaningful variance.
- **Greedy assignment, not Hungarian.** Optimal for ≤5 clauses in practice, but not guaranteed.
- **Region boxes come from annotations.** At 1M images this needs a detector (LookSync uses SAM v2 + Florence as its fallback path), which introduces detector error the current numbers do not include.

6. Scalability to 1M images

Candidate generation is an ANN lookup in ChromaDB over the whole corpus; only the shortlist is scored densely. Cost is O(log N) in corpus size, so the retrieval logic is unchanged at 1M images. Three real costs:

- **Index size grows ~3.4×** (6,036 vectors for 1,788 images). At 1M images that is ~3.4M vectors × 768 dims × 4 bytes ≈ 10 GB — past comfortable RAM. Fix: product quantisation, or index regions only for garment-bearing images.
- **Dense scoring is O(candidates × clauses × regions)**, but candidates are capped at ~150/clause, so this is constant in N.
- **Chroma is the right call at this scale and the wrong one at 1M.** The brief says to pick the convenient store; the migration path is Qdrant or Vertex AI Vector Search (LookSync serves 18M+ products at P90 < 1s with an offline-index / online-serve split plus aggressive caching).

7. Future work

(a) Adding locations (cities, places) and weather

The architecture already has the right seam: **scene clauses route to the global vector**, so location and weather are a third clause role rather than a redesign.

- **Add a role, not a model.** Extend the anchor set with 'a city or geographic location' and 'the weather or time of day'. Routing is similarity-based, so this is additive — no retraining, no vocabulary.
- **Cheap metadata first.** EXIF GPS and capture time give city and season for free where present; both become Chroma metadata filters, which cut the candidate set *before* any vector work and so cost nothing at query time.
- **Where labels are absent, use a dedicated encoder.** A geo-aware model (StreetCLIP was trained for exactly this) tags city/region; weather is well within zero-shot CLIP ('an overcast day', 'rain-soaked street'). Store as a fourth vector kind + metadata.
- **Weather is a garment prior, not just a filter** — the commercially interesting part. 'What to wear in Bangalore in July' is a monsoon query: it should surface raincoats and closed shoes. That is a learned correlation between weather and garment category, not a retrieval constraint, and it is where a recommendation layer would sit on top of this retriever.

(b) Improving precision

- **Cross-encoder rerank (approach D) is the biggest lever left.** BLIP-ITM over the top-50: cross-attention resolves binding properly rather than approximating it with assignment. Confined to a shortlist it is affordable, and the architecture already produces that shortlist. This is the first thing I would build next.
- **Hard-negative mining.** Colour-swapped and garment-swapped negatives are exactly what the swap metric measures — mine them automatically from the corpus and fine-tune (approach E). This pushes binding into the embedding so it survives at ANN time instead of being reconstructed at scoring time.
- **Replace the syntactic splitter with an LLM decomposer**, as LookSync does. It fixes the query-4 failure directly and handles possessives, negation ('not black') and implicit garments ('business attire' → blazer + button-down) that regex cannot.
- **Learn the fusion weights.** W_PRIOR and the clause weights are hand-set. With even a few hundred judged pairs they should be fit, not guessed.
- **Detector-based regions.** Removes the annotation dependency and lets region scoring work on arbitrary images — required for this to run on real traffic.

8. Codebase

https://github.com/<your-username>/glance-fashion-retrieval

`data/` corpus construction + audits · `indexer/` Part A: encoders, crops, vector storage · `retriever/` Part B: decomposition, routing, assignment, fusion · `eval/` graded queries + stress test. Logic is separated from data throughout: the corpus is JSON + images, every model is swappable by a flag, and the eval harness runs any backbone/config combination without touching retrieval code.